Natalia Isabel Martínez Gómez

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Physics student at the Universidad del Quindío (Armenia, Colombia), currently working on my thesis in neutrino physics. I am interested in particle physics, astroparticle physics and computational methods.

RESEARCH

Neutrino mass and sterile neutrinos with KATRIN data Thesis (in progress)

Analysis of the first KATRIN science run (KNM1) in Fortran: fit of the effective electron-neutrino mass squared, m_β^2, and exclusion contours for a 3+1 sterile-neutrino scenario.

EDUCATION

Physics — Universidad del Quindío, Armenia 2020 – present

All coursework completed; thesis in progress.

Soil Laboratory Technician — SENA Nariño, Pasto 2018 – 2020

ACADEMIC EXPERIENCE

Teaching Assistant — Universidad del Quindío, Armenia 08/2023 – 05/2025

CONFERENCES

XXX Congreso Nacional de Física (Colombian National Physics Congress) — Universidad Nacional de Colombia, Manizales Oct 8–10, 2025

Poster: *"Analysis of the tritium beta-decay spectrum in KATRIN to set limits on sterile-neutrino parameters"*

9th Colombian Meeting on High Energy Physics (ComHEP) — Universidad de Nariño, Pasto Dec 2–6, 2024

Attendee

XXIX Congreso Nacional de Física (with the Sociedad Colombiana de Física) — Universidad del Quindío, Armenia Sep 21–23, 2022

Attendee

TOOLS

Python · Fortran · Jupyter · Git

LANGUAGES

Spanish (native) · English (B1)